IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-136

1. Topic (12 words max)

Aegis Life: Integrated AI Platform for Holistic Wellness

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	No
Year	No

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka currently faces a fragmented digital healthcare environment in which nutrition management, physical fitness tracking, medical monitoring, and mental wellness support are delivered through isolated, non-communicating platforms. This fragmentation prevents individuals from obtaining a comprehensive and continuous view of their overall health, forcing users to rely on multiple applications or manual tracking methods. As a result, health-related decisions tend to be reactive, inconsistent, and poorly informed, thereby reducing the effectiveness of preventive healthcare and long-term lifestyle management (World Health Organization [WHO], 2022).

The severity of this problem affects multiple population groups. Working professionals aged 25–45 increasingly experience sedentary lifestyles combined with chronic conditions such as diabetes and hypertension, yet lack integrated digital support tailored to their daily routines. University students face high academic and work-related stress alongside irregular dietary habits, negatively impacting both mental and physical well-being. Middle-aged adults seeking preventive healthcare lack coordinated guidance across nutrition, fitness, and medical monitoring, while fitness enthusiasts depend on generic workout and diet plans that fail to provide personalized, data-driven integration across lifestyle domains (International Diabetes Federation [IDF], 2021).

The impact of this fragmented health ecosystem is evident at social, economic, technical, ethical, and legal levels. Socially, Sri Lanka has a high prevalence of lifestyle-related non-communicable diseases, with diabetes affecting approximately 27% of the adult population, a situation exacerbated by poor lifestyle integration and limited preventive care mechanisms (IDF, 2021).

Economically, national healthcare expenditure continues to increase annually, placing a substantial burden on individuals and the public healthcare system due to largely preventable conditions (WHO, 2022).

From a technical perspective, the lack of localized digital health and artificial intelligence solutions that accommodate Sri Lankan dietary habits, medical report formats, and culturally specific stress patterns significantly reduces the effectiveness of existing platforms. Ethically and legally, health data dispersed across multiple applications raises serious concerns related to privacy, data security, and the absence of unified, compliant health data governance frameworks. These challenges collectively justify the need for an integrated, culturally adapted digital healthcare solution.

This research is positioned within relevant Focus of Competence (FoC) research clusters, including Microservices Architecture Design, Distributed Data Management, System Integration Patterns, and Mobile Platform Optimization. Furthermore, it contributes to multiple industry verticals such as Healthcare Technology, Wellness and Fitness, Pharmaceutical and Dietary Supplements, and Telemedicine, highlighting its interdisciplinary and applied research value.

The identified research problem directly aligns with the United Nations Sustainable Development Goals, particularly SDG 3: Good Health and Well-being, by promoting preventive healthcare, early detection of health risks, and continuous lifestyle management through integrated digital solutions. By enabling individuals to access coordinated insights across nutrition, fitness, mental wellness, and medical monitoring, the proposed approach supports healthier lifestyles and reduces the long-term burden of non-communicable diseases (United Nations, 2015; WHO, 2022). Additionally, the research contributes to SDG 9: Industry, Innovation, and Infrastructure by fostering innovation in digital health system design, and SDG 10: Reduced Inequalities by improving access to comprehensive health information for diverse population groups, including students, professionals, and middle-aged adults, regardless of socioeconomic background (United Nations,2015).

References

- International Diabetes Federation. (2021). *IDF diabetes atlas* (10th ed.). International Diabetes Federation.
- United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development*. United Nations.
- World Health Organization. (2022). *Global strategy on digital health 2020–2025*. World Health Organization.

6. Brief description of Existing Research and Systems (300 words max)

Existing **nutrition and dietary management systems** such as *MyFitnessPal* provide extensive food databases and calorie tracking features; however, these databases are primarily designed for Western cuisines and do not support Sri Lankan food items or local portion styles. Research on AI-based food recognition has shown promising results in accurately identifying food items and estimating nutritional values, but these models are largely trained on Western datasets (Chen et al., 2018). As a result, current systems fail to provide accurate dietary recommendations for Sri Lankan users, limiting their effectiveness in local contexts.

In the area of **fitness and physical activity tracking**, platforms like *Nike Training Club* offer structured workout plans and general fitness guidance. However, these systems rely on generic routines without considering users' dietary intake, medical conditions, or local lifestyle constraints. Research on pose estimation and exercise monitoring demonstrates that AI-based analysis can effectively detect exercise form and performance (Cao et al., 2019). Despite this, existing applications do not integrate fitness data with nutrition or health metrics, reducing personalization and increasing the risk of ineffective or unsafe exercise practices.

Regarding **medical monitoring and health data management**, patient portals such as *MyChart* focus primarily on appointment scheduling and basic medical record access. These systems lack advanced analytics and lifestyle integration, offering limited support for preventive healthcare. Integrated digital health platforms have been shown to significantly improve health outcomes by enabling coordinated care and continuous monitoring (Kumar et al., 2019). However, current solutions do not support Sri Lankan lab report formats or localized medical data processing, creating barriers to adoption and automation.

For **mental health and stress management**, applications like *Headspace* provide guided meditation and mindfulness content based on Western stress models. Research indicates that facial analysis and AI-based stress detection techniques are effective in identifying stress levels (Saeed et al., 2021), and culturally adapted interventions significantly improve user adherence in South Asian populations (Patel et al., 2020). Nevertheless, existing systems lack cultural adaptation to Sri Lankan stress patterns and operate independently of physical health and lifestyle data, limiting their overall impact.

References

1. Kumar, A., et al. (2019). *Impact of integrated digital health platforms on health outcomes.* Journal of Medical Systems, 43(2), 45.
2. Chen, X., et al. (2018). *Food recognition using AI for dietary management.* IEEE Transactions on Multimedia, 20(9), 2458–2469.
3. Cao, Z., et al. (2019). *Real-time pose estimation for exercise monitoring.* International Journal of Computer Vision, 127(3), 319–338.
4. Saeed, F., et al. (2021). *Facial analysis for stress detection in digital health applications.* Computers in Human Behavior, 115, 106609.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

This solution addresses Sri Lanka's fragmented digital health landscape by introducing "Aegis Life", an integrated AI-powered wellness ecosystem. The platform unifies four specialized modules NutriAI, FitAI, VitalAI, and MindAI into a single, culturally-adapted mobile application, bridging gaps in existing systems related to integration, localization, and user adherence. By leveraging AI models specifically trained on Sri Lankan food items, local medical lab formats, and regional stress patterns, the system ensures that preventive and personalized healthcare is both accessible and contextually relevant.

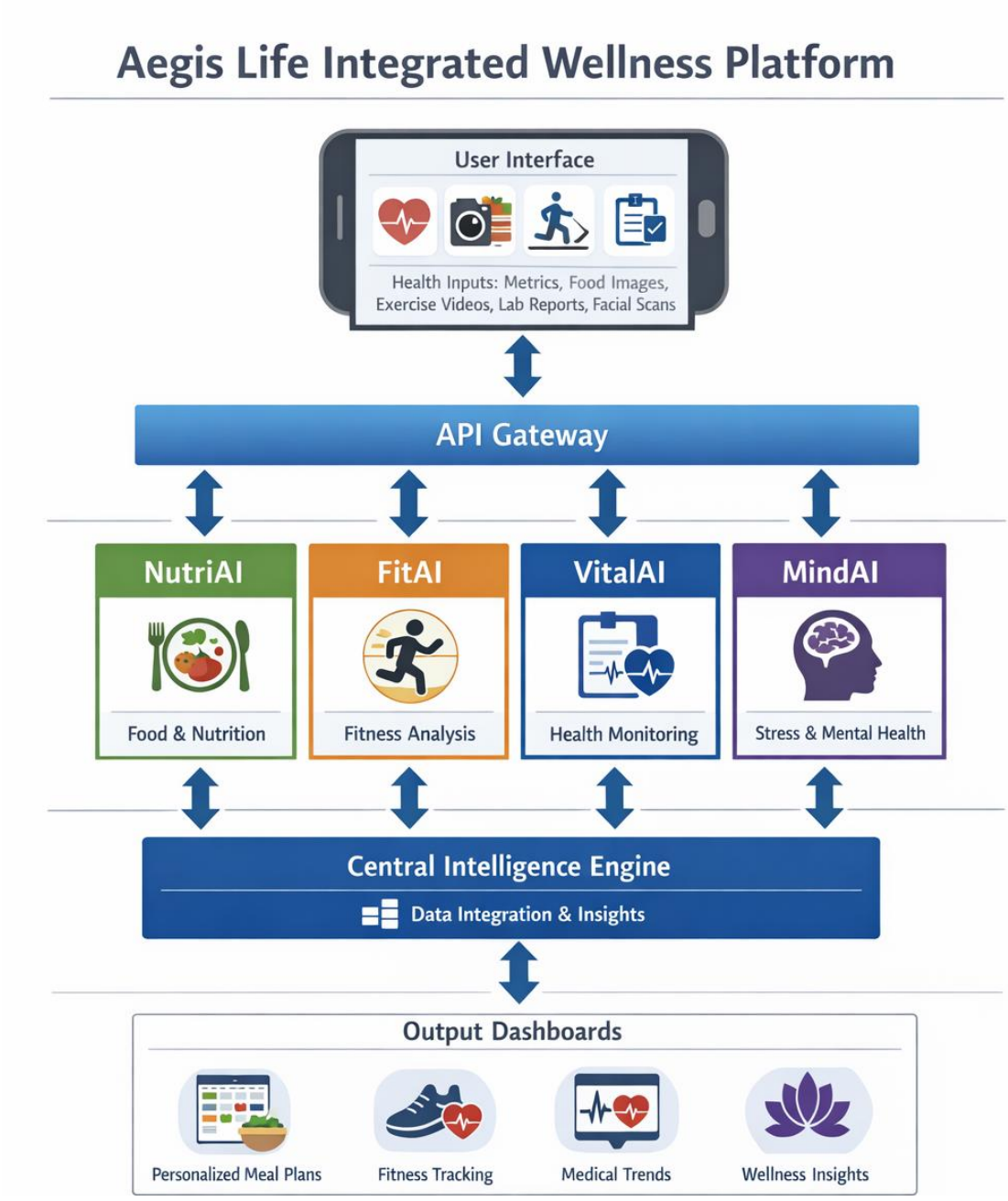
NutriAI focuses on dietary management and resolves the lack of localized food recognition in global applications. Using AI-powered food image recognition, NutriAI identifies traditional Sri Lankan dishes such as kottu, hoppers, and rice & curry, estimates their nutritional values, and integrates them with users' health metrics. It also cross-references supplement and medical data to ensure safe recommendations. By doing so, NutriAI addresses the gap in local dietary guidance while promoting responsible consumption and healthier lifestyle choices.

FitAI provides personalized fitness guidance and overcomes the limitations of generic workout apps. Using pose estimation and activity recognition, it analyzes exercise videos and tracks form in real time. The module adapts routines to local lifestyles and resource constraints, incorporating offline capabilities for users with limited connectivity. FitAI bridges the integration gap between diet, fitness, and health monitoring, ensuring that users receive holistic guidance that aligns with their personal goals and environmental conditions.

VitalAI tackles the challenges associated with medical data integration and monitoring. It processes lab reports from local hospitals using OCR, analyzes vital health metrics, and identifies trends to support early detection and preventive care. By integrating with NutriAI and FitAI, VitalAI ensures that medical insights are considered alongside lifestyle interventions, enabling safe, informed, and personalized health decisions.

MindAI addresses mental health and stress management through culturally adapted AI. It analyzes facial expressions, daily routines, and user-reported data to detect stress patterns specific to Sri Lankan contexts, such as exam pressures or work-related stress. MindAI generates personalized recommendations, including mindfulness exercises and wellness strategies, ensuring mental well-being is addressed alongside physical health. This integration fosters adherence to preventive care while providing a complete wellness experience within a single cohesive platform.

Overall, the Aegis Life architecture employs a layered approach where a unified user interface feeds into an API gateway orchestrating all four modules. This design enables seamless data exchange, real-time insights, modular scalability, and strict data privacy, resulting in a fully integrated, culturally relevant, and preventive healthcare ecosystem for Sri Lankan users.


Aegis Life Integrated Wellness Platform
User Interface
Health Inputs: Metrics, Food Images, Exercise Videos, Lab Reports, Facial Scans
API Gateway
NutriAI
Food & Nutrition
FitAI
Fitness Analysis
VitalAI
Health Monitoring
MindAI
Stress & Mental Health
Central Intelligence Engine
Data Integration & Insights
Output Dashboards
Personalized Meal Plans
Fitness Tracking
Medical Trends
Wellness Insights

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful development of Aegis Life relies heavily on access to specialized domain expertise and knowledge across nutrition, fitness, medical, and mental health domains. The project team leverages academic resources such as the SLIIT library, as well as industry partnerships with local hospitals, fitness centers, and Ayurvedic practitioners to gain practical, context-specific insights. Expert consultations with nutritionists and fitness trainers further enhance the team's understanding of local dietary habits, exercise techniques, and preventive healthcare measures. Field research, including surveys and observations within Sri Lankan communities, provides invaluable information about cultural wellness practices and stress management strategies, ensuring the solution is both relevant and culturally sensitive. Online learning platforms such as YouTube and Wikipedia supplement these efforts by keeping the team updated on emerging methodologies and technologies in digital health systems.

High-quality, localized datasets are critical to developing effective AI models and addressing the limitations of existing systems. The project requires comprehensive datasets, including Sri Lankan food images with nutritional information to train NutriAI, medical report formats from local hospitals for VitalAI, and exercise video demonstrations reflecting local conditions for FitAI. MindAI utilizes facial expression datasets capturing stress patterns across Sri Lankan demographics, while a health supplement database provides verified information on locally available supplements. Additionally, anonymized user health and lifestyle data, covering dietary patterns, exercise habits, and wellness routines, enable the creation of personalized and contextually appropriate recommendations. Cultural wellness datasets, including traditional remedies and local stress reduction practices, further strengthen the system's relevance and adherence potential. These datasets collectively ensure that AI models are trained with accurate, culturally adapted, and comprehensive information, overcoming the shortcomings of generic global solutions.

Ethical oversight and data governance form another essential pillar of the project. Due to the sensitive nature of health, psychological, and lifestyle data, the project will obtain ethical clearance from the SLIIT Ethics Committee. All data collection follows strict protocols, including informed consent in local languages, complete anonymization, encrypted storage, regular privacy audits, and clear medical disclaimers. This ensures that the system respects user privacy while maintaining data integrity. By combining domain expertise, specialized knowledge, and high-quality localized data under stringent ethical guidelines, the team can address the gaps identified in current digital health solutions and develop an integrated, AI-driven wellness ecosystem that is safe, personalized, and culturally relevant for Sri Lankan users.

9. Objectives and Novelty

Main Objective:

To develop an integrated AI-powered health platform that provides personalized wellness management through four interconnected modules: nutrition tracking, fitness guidance, medical monitoring, and mental wellness support, specifically adapted for the Sri Lankan context.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Saisuhani L - IT22091598	To develop Sri Lankan Food Recognition & Nutrition Management System that accurately identifies local foods and provides culturallyappropriate dietary recommendations.	• Fine-tune CNN models for Sri Lankan food image recognition • Develop local food nutritional database • Create budget-aware meal planning algorithm • Integrate with supplement recommendation system	1. Localization of AI Models: This is the first AI-powered dietary system trained specifically on Sri Lankan cuisine, overcoming the inability of global apps to recognize local foods. 2. Budget-aware Meal Planning: Unlike existing systems, NutriAI considers users' financial constraints, making healthy eating both practical and sustainable in the local context. 3. Integration with Local Supplements: The system links meal plans with verified local supplement data, providing automated, safe recommendations that prevent self-prescription errors. 4. Innovative Methods and Algorithms: The module leverages CNN-based image recognition for food identification,

			optimization algorithms for meal planning under budget constraints, and data cross-referencing techniques for supplement safety integration. 5. Enhanced User Experience and Accuracy: By providing culturally adapted food recognition, personalized and affordable meal plans, and seamless integration with supplements, the system significantly improves usability, accuracy, and decision-making efficiency. 6. Intelligent Data Utilization: NutriAI uses diverse datasets, including local food images, nutritional information, and supplement data, to generate intelligent, real-time dietary recommendations. 7. Overall Advantage Over Existing Systems: This module solves critical gaps in current health apps lack of local food recognition, no integration of nutrition with supplements, and absence of budget-conscious recommendations providing a holistic, personalized, and contextually relevant dietary management solution for Sri Lankan users.
Wijesooriya W.M.R.A.B - IT22249098	To develop an AI-Powered Adaptive Fitness Ecosystem that creates personalized workout schedules by integrating data from	• Design adaptive workout scheduler based on NutriAI calorie intake and goals • Develop home/gym workout plans with GIF demonstrations	1. Dynamic Personalization: This is the first integrated fitness system in Sri Lanka that adjusts workouts in real time based on nutrition, medical conditions, and mental state, ensuring safe, effective, and personalized guidance.

	all modules and enabling social fitness challenges and with local business integration.	• Create medical conditionaware exercise modifications (diabetes, hypertension) • Implement MindAIpowered motivational system with goal tracking • Build social challenge platform for family/friend fitness competitions • Develop local supplement brand marketplace with targeted advertising • Create achievement ranking system with Sri Lankan cultural elements	2. Integration with Nutrition and Mental Health: By combining data from NutriAI and MindAI, FitAI creates a holistic approach, bridging gaps in existing fitness apps that operate in isolation. 3. Innovative Methods and Algorithms: The module leverages adaptive scheduling algorithms, exercise modification logic for health conditions, and motivational analytics driven by AI. GIF demonstrations enhance clarity and correctness of workouts. 4. Enhanced User Experience: Users receive culturally relevant guidance, visual aids, and motivational feedback, improving adherence, engagement, and performance accuracy. 5. Intelligent Data Usage and Automation: FitAI utilizes real-time nutrition, fitness, mental health, and medical data to generate automatic adjustments and recommendations, reducing the need for manual planning. 6. Social and Gamification Features: Family/friend challenges, achievement ranking with cultural relevance, and supplement marketplace integration create a sustainable, engaging ecosystem not present in conventional apps. 7. Improvement Over Existing Systems: Unlike generic global fitness apps, FitAI offers a personalized, culturally adapted, integrated, and interactive fitness platform, addressing

			gaps in exercise customization, health-aware modifications, engagement, and local relevance.
Dhananjaya A.G.G - IT22145734	To develop Sri Lankan Medical Report Parser & Health Monitoring System that digitizes and analyzes local lab reports for integrated health insights.	• Train OCR model for Sri Lankan lab report formats • Develop medical data extraction and trend analysis • Implement supplement safety checking system Create health risk alert mechanisms	1. Localization of OCR Technology: This is the first OCR system trained specifically on Sri Lankan lab report formats, addressing the gap where global systems fail to recognize local documentation styles. 2. Integration with Lifestyle Data: VitalAI cross-references lab results with nutrition (NutriAI) and fitness (FitAI) data, providing a holistic understanding of user health, which existing patient portals do not offer. 3. Innovative Methods: The module combines OCR, automated trend analysis, and drug-supplement interaction algorithms, providing intelligent, real-time insights and recommendations. 4. Enhanced Accuracy and Efficiency: Automated extraction and analysis of lab data reduce manual effort, increase accuracy, and deliver actionable insights quickly. 5. New Data Usage: The system uses anonymized lab reports, medical histories, and supplement data to generate personalized safety alerts, enhancing preventive care. 6. Improved User Experience: Users receive a unified view of their medical trends, automated alerts, and safety guidance,

			making health management safer and more convenient. 7. Advantage Over Existing Systems: Unlike standard patient portals or global apps, VitalAI addresses fragmented healthcare by integrating medical data with lifestyle and dietary information, ensuring personalized, culturally relevant, and preventive health management for Sri Lankan users.
Gunathilake D.R.G.L-IT22544636	To develop Culturally-Aware Mental Wellness System with Facial Stress Detection adapted for Sri Lankan demographics and stress patterns.	• Train facial stress detection model on Sri Lankan facial features • Develop exam/work stress pattern recognition • Create integration with fitness goal achievement motivation • Implement traditional Ayurvedic remedy suggestions	1. Culturally Adapted Stress Detection: MindAI is the first AI system calibrated for Sri Lankan facial features and stress patterns, addressing the limitations of global meditation or stress apps that are based on Western populations. 2. Integration with Lifestyle and Fitness Data: By connecting stress detection to fitness goals (FitAI) and nutrition data (NutriAI), the module delivers holistic, personalized wellness recommendations, filling the gap of isolated mental health apps. 3. Innovative Methods: Combines facial recognition algorithms, behavioral pattern analysis, and culturally specific stress modeling. Traditional wellness practices, such as Ayurvedic remedies, are incorporated algorithmically for automated suggestions. 4. Enhanced User Experience and Accuracy: Users receive real-time, actionable stress insights and culturally relevant wellness

			recommendations, improving engagement, adherence, and overall well-being. 5. New Data Usage and Intelligence: Utilizes facial datasets, lifestyle metrics, and local stress surveys to generate AI-driven insights, adding intelligence that existing solutions lack. 6. Efficiency and Automation: Automated detection of stress triggers and integration with preventive interventions reduces manual tracking and promotes proactive mental health management. 7. Advantage Over Existing Solutions: Unlike global meditation or generic stress management apps, MindAI offers a personalized, culturally sensitive, and fully integrated mental health solution, enhancing preventive care and overall wellness for Sri Lankan users.

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Kaushalya	Dissanayake	K. Dissanayake
Co-Supervisor	Ms.	Suriya	Kumari	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				